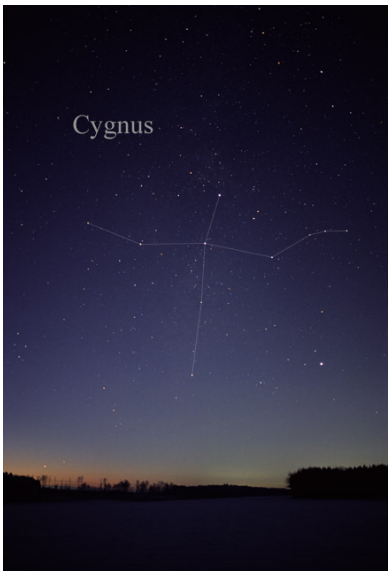
Assignment. In this project, we will look at astrological predictions that the ancients could only dream of. The basis for our astronomical calculations will be the python library `astropy` and possibly `sunpy`. The former can be used to compute coordinates of celestial bodies in numerous coordinate systems as well as compute distances in those coordinates. As an example, we will take two bright stars visible in the night sky, Deneb and Vega. Deneb is found at the end of the short arm (the swan's butt) of Cygnus, while Vega is at the far corner of Lyra (the g string of a lyre, imagination required.)

The stars are relatively close in the night sky, see if you can spot one constellation in the photo of the other below.





Astropy knows the names of these stars as well as their Bayer classifications, α Cyg and α Lyr. For any given night sky at any given latitude and longitude, astropy can calculate the locations of both stars. We'll use the altitude/azimuth coordinates; that's (viewing angle, angle from magnetic north). Let's go back to June 2025 and find the coordinates and their angular separation for the first of each month until February 2026 as viewed in the night sky at coordinates 20.68298° N, 88.56867° W (put those in Google Maps to see the location). We find the following angular separations measured in degrees:

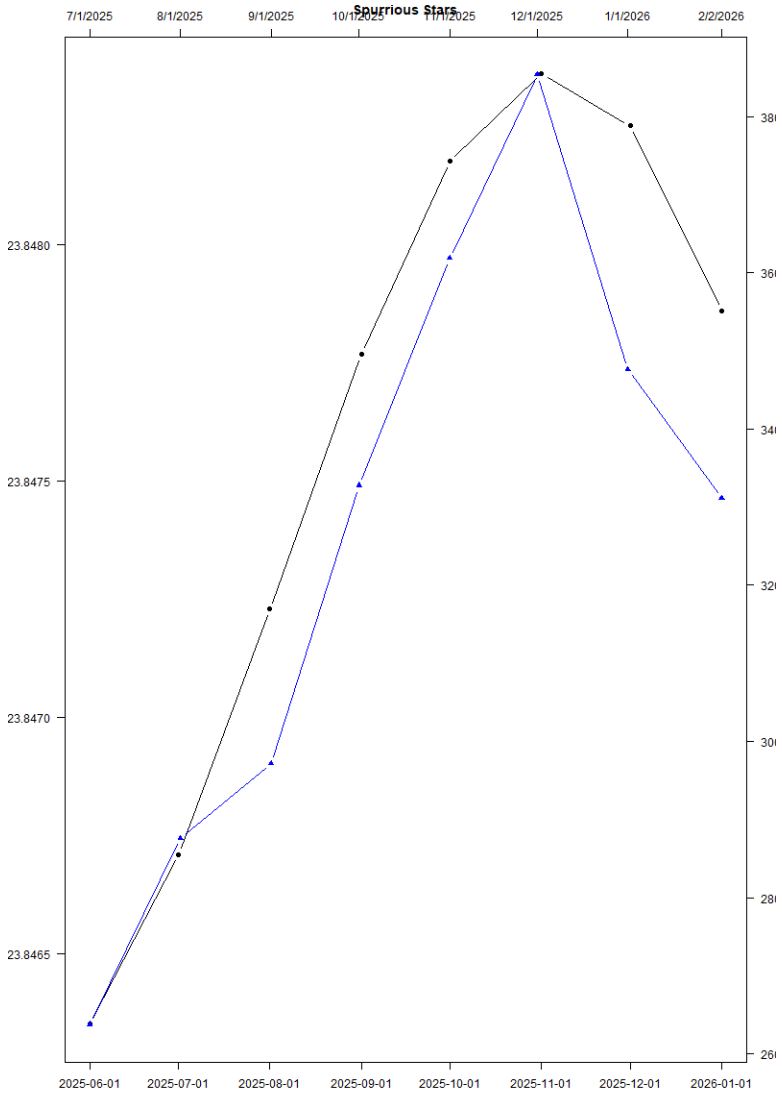
Date	Deg Sep
2025-06-01	23.84634901
2025-07-01	23.84670648
2025-08-01	23.84722538
2025-09-01	23.84776596
2025-10-01	23.84817564
2025-11-10	23.84836234
2025-12-01	23.84825318
2026-01-01	23.8478633
2026-02-01	23.84730532

Our goal now is to correlate the above time-series with some other data; let's go financial. After some digging, the stock AVGO (that's Broadcom) had the following average closing prices for the months of June 2025 to February 2026.

Date	Avg Price
2025-06	247.228546
2025-07	263.782440
2025-08	287.596039
2025-09	297.161316
2025-10	332.753326
2025-11	361.857605
2025-12	385.342682
2026-01	347.619995
2026-02	331.109985

Pairing the separations of Deneb and Vega with the average monthly closing prices of Broadcom's stock, we compute a Pearson's correlation coefficient of $r = .896$, quite strong. Because our computations are for nonsense anyway, we shouldn't be afraid to shift dates around or rescale at will. If we drop the most recent date from the star distances and the earliest date from the average prices we get a correlation coefficient of $r = 0.964$. This is the range we will look for, ensuring we get visually impressive graphs. One could interpret the shift in the dates as the change in separation between the stars is predicting the change in stock price for the next month. Hmmm, useful if it were realistic.

We can overlay the line graphs of these two data-sets. Because we have shifted the dates, we will want sets of x and y-axes for both sets of data. That's four axes. We get something like:



Back to the Assignment. We want to perform these types of correlations on a large scale, creating csv's of various astronomical computations (any that come to mind) and companion sets of real world data: stock-prices, agricultural data, etc. The aim is to have high correlation, $r^2 \approx 1$ creating closely matching graphs. The above example can be easily out-done.